

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

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Experiment 2

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Branch: B.E. CSE

Semester: 5th

Subject Name: Full Stack

UID: 24BCS11074

Section/Group: 610-B

Date of Performance: July 2026

Subject Code: 24CSP-304

Easy

1. **Aim:** Build a counter app with increment, decrement, and reset buttons using `useState`. Display the live count and apply colour change when count is negative.

2.Objective:

- Implement a simple counter using `useState`.
- Add buttons for **increment, decrement, and reset** actions.
- Display the live count dynamically.
- Apply **conditional styling** (colour change when count is negative).

3.Code:

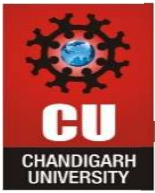
- `index.css`

```
body {  
  margin: 0;  
  font-family: 'Courier New', Courier, monospace;  
  background-color: aliceblue;  
}
```

```
.container {  
  text-align: center;  
  margin-top: 50px;  
}
```

```
button {  
  padding: 10px 20px;  
  margin: 10px;  
  font-size: large;  
  cursor: pointer; /* fixed */  
  border: none;  
  border-radius: 5px;  
}
```

```
button:hover {  
  background-color: aqua;  
}
```



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- CounterApp.jsx

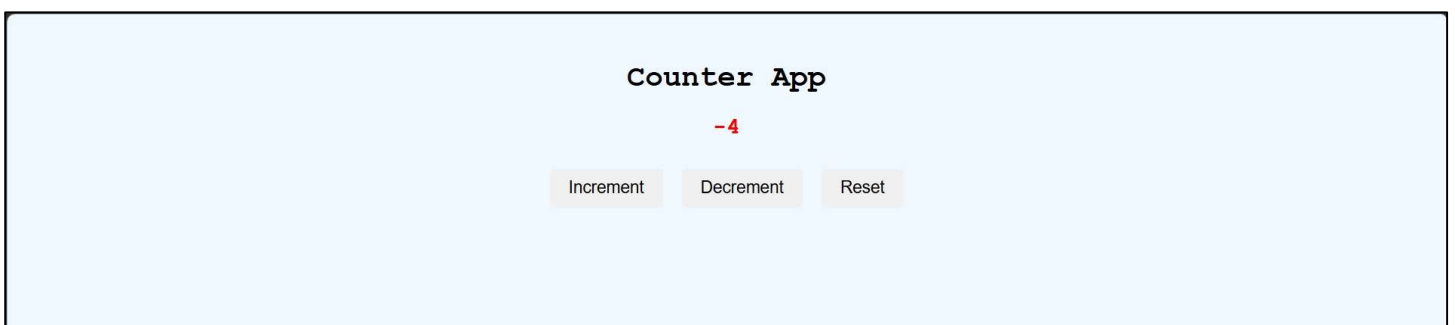
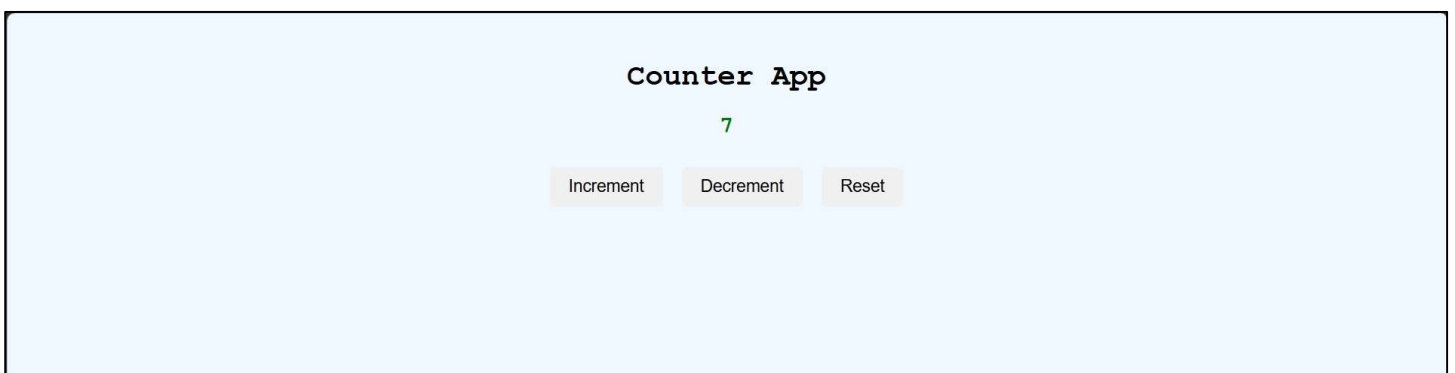
```
import React, { useState } from "react";
import "./index.css";
function App() {
  const [count, setCount] = useState(0);

  const increment = () => setCount(count + 1);
  const decrement = () => setCount(count - 1);
  const reset = () => setCount(0);

  return (
    <div className="container">
      <h1>Counter App</h1>
      <h2 style={{ color: count < 0 ? "red" : "green" }}>
        {count}
      </h2>
      <div>
        <button onClick={increment}>Increment</button>
        <button onClick={decrement}>Decrement</button>
        <button onClick={reset}>Reset</button>
      </div>
    </div>
  );
}

export default App;
```

4. Output:



5. Learning Outcome:

- **Understanding useState** for state management in React.
- **Ability to handle events** with button clicks.
- **Confidence in conditional rendering** for UI changes.
- **Skill in applying inline styles** based on state values.
- **Foundation for interactive React apps** with dynamic behaviour.

Medium

1. **Aim:** Create a controlled student registration form with fields for name, email, course, and age using useState for two-way binding. Display submitted data below the form without page reload.

2.Objective:

- Create a controlled form with fields (name, age, email, course).
- Use **useState** for two-way binding of input values.
- Handle form submission without page reload.
- Display submitted data dynamically below the form.

3.Code:

- App.jsx

```
import React, { useState } from "react";
import "./index.css";

function App() {
  const [formData, setFormData] = useState({
    name: "",
    age: "",
    email: "",
    course: ""
  });

  const handleChange = (e) => {
    const { name, value } = e.target;
    setFormData({ ...formData, [name]: value });
  };

  const handleSubmit = (e) => {
    e.preventDefault(); // prevent reload
    console.log("Form submitted:", formData);
  };
}
```



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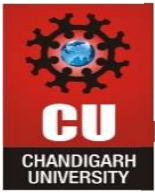
```

return (
<div className="container">
  <h1>School Registration Form</h1>
  <form onSubmit={handleSubmit}>
    <input
      type="text"
      name="name"
      placeholder="Enter Name"
      value={formData.name}
      onChange={handleChange}
    />
    <input
      type="number"
      name="age"
      placeholder="Enter Age"
      value={formData.age}
      onChange={handleChange}
    />
    <input
      type="email"
      name="email"
      placeholder="Enter Email"
      value={formData.email}
      onChange={handleChange}
    />
    <input
      type="text"
      name="course"
      placeholder="Enter Course"
      value={formData.course}
      onChange={handleChange}
    />
    <button type="submit">Register</button>
  </form>

  <div className="output">
    <h2>Entered Details:</h2>
    <p><b>Name:</b> {formData.name}</p>
    <p><b>Age:</b> {formData.age}</p>
    <p><b>Email:</b> {formData.email}</p>
    <p><b>Course:</b> {formData.course}</p>
  </div>
</div>
);
}

```

```
export default App;
```



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- **index.css**

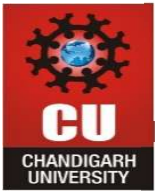
```
body {  
  
    margin: 0;  
  
    font-family: 'Courier New', Courier, monospace;  
  
    background-color: aliceblue;  
  
}
```

```
.container {  
  
    text-align: center;  
  
    margin-top: 50px;  
  
}
```

```
input {  
  
    display: block;  
  
    margin: 10px auto;  
  
    padding: 8px;  
  
    font-size: 16px;  
  
}
```

```
button {  
  
    padding: 10px 20px;  
  
    margin-top: 10px;  
  
    font-size: 16px;  
  
    cursor: pointer;  
  
}
```

```
button:hover {
```



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```
background-color: aqua;  
}  
  
.output {  
margin-top: 20px;  
}
```

4. Output:

School Registration Form

Yash Sharma

21

yashsharma56@outlook.co

B.E. CSE

Register

Entered Details:

Name: Yash Sharma

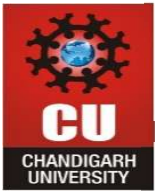
Age: 21

Email: yashsharma56@outlook.com

Course: B.E. CSE

5. Learning Outcome:

- Understanding controlled components in React.
- Ability to bind form inputs with state for live updates.
- Skill in handling form events using `onChange` and `onSubmit`.
- Confidence in preventing default browser behaviour (no reload).
- Capability to display user input dynamically in the UI.



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Hard

1. **Aim:** Implement a student data table that fetches records from a public REST API (JSONPlaceholder) using `useEffect` on mount, shows a loading spinner during fetch, handles errors gracefully, and supports live search filtering by name.

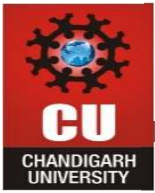
2.Objective:

- Fetch student records from a public REST API using **`useEffect`**.
- Show a **loading spinner** while data is being fetched.
- Handle errors gracefully with proper messages.
- Implement **live search filtering** by student name.
- Render data in a structured **table format**.

3.Code:

- **index.css**

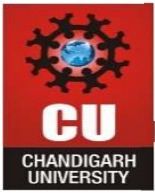
```
.counter {  
  
  font-size: 16px;  
  
  padding: 5px 10px;  
  
  border-radius: 5px;  
  
  color: var(--accent);  
  
  background: var(--accent-bg);  
  
  border: 2px solid transparent;  
  
  transition: border-color 0.3s;  
  
  margin-bottom: 24px;  
  
  &:hover {  
  
    border-color: var(--accent-border);  
  
  }  
  
  &:focus-visible {  
  
    outline: 2px solid var(--accent);  
  
    outline-offset: 2px;  
  
  }  
}
```



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```
}  
  
}  
  
.hero {  
  
    position: relative;  
  
  
  
    .base,  
  
    .framework,  
  
    .vite {  
  
        inset-inline: 0;  
  
        margin: 0 auto;  
  
    }  
  
  
  
    .base {  
  
        width: 170px;  
  
        position: relative;  
  
        z-index: 0;  
  
    }  
  
  
  
    .framework,  
  
    .vite {  
  
        position: absolute;  
  
    }  
  
  
  
    .framework {  
  
        z-index: 1;  
  
        top: 34px;  
  
        height: 28px;
```

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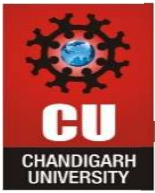
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```
transform: perspective(2000px) rotateZ(300deg) rotateX(44deg) rotateY(39deg)
    scale(1.4);
}

.vite {
    z-index: 0;
    top: 107px;
    height: 26px;
    width: auto;
    transform: perspective(2000px) rotateZ(300deg) rotateX(40deg) rotateY(39deg)
        scale(0.8);
}

#center {
    display: flex;
    flex-direction: column;
    gap: 25px;
    place-content: center;
    place-items: center;
    flex-grow: 1;

    @media (max-width: 1024px) {
        padding: 32px 20px 24px;
        gap: 18px;
    }
}
```



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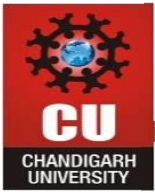
```
#next-steps {  
  display: flex;  
  border-top: 1px solid var(--border);  
  text-align: left;
```

```
& > div {  
  flex: 1 1 0;  
  padding: 32px;  
  @media (max-width: 1024px) {  
    padding: 24px 20px;  
  }  
}
```

```
.icon {  
  margin-bottom: 16px;  
  width: 22px;  
  height: 22px;  
}
```

```
@media (max-width: 1024px) {  
  flex-direction: column;  
  text-align: center;  
}  
}
```

```
#docs {
```



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```
border-right: 1px solid var(--border);

@media (max-width: 1024px) {

  border-right: none;

  border-bottom: 1px solid var(--border);

}

}
```

```
#next-steps ul {

  list-style: none;

  padding: 0;

  display: flex;

  gap: 8px;

  margin: 32px 0 0;
```

```
.logo {

  height: 18px;

}
```

```
a {

  color: var(--text-h);

  font-size: 16px;

  border-radius: 6px;

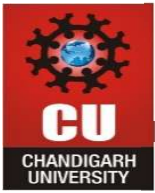
  background: var(--social-bg);

  display: flex;

  padding: 6px 12px;

  align-items: center;

  gap: 8px;
```



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text-decoration: none;

transition: box-shadow 0.3s;

&:hover {

box-shadow: var(--shadow);

}

.button-icon {

height: 18px;

width: 18px;

}

}

@media (max-width: 1024px) {

margin-top: 20px;

flex-wrap: wrap;

justify-content: center;

li {

flex: 1 1 calc(50% - 8px);

}

a {

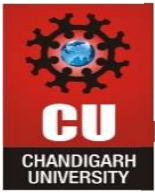
width: 100%;

justify-content: center;

box-sizing: border-box;

}

}



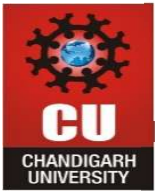
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```
}  
  
#spacer {  
  
    height: 88px;  
  
    border-top: 1px solid var(--border);  
  
    @media (max-width: 1024px) {  
  
        height: 48px;  
  
    }  
}
```

```
.ticks {  
  
    position: relative;  
  
    width: 100%;  
  
    &::before,  
    &::after {  
  
        content: " ";  
  
        position: absolute;  
  
        top: -4.5px;  
  
        border: 5px solid transparent;  
  
    }
```

```
&::before {  
  
    left: 0;  
  
    border-left-color: var(--border);  
  
}  
  
&::after {  
  
    right: 0;
```



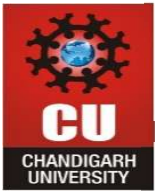
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```
border-right-color: var(--border);  
  
}  
  
}
```

- **App.jsx**

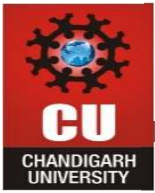
```
import React, { useState, useEffect } from "react";  
  
import StudentTable from "./StudentTable";  
  
import "./index.css";  
  
function App() {  
  
  const [students, setStudents] = useState([]);  
  
  const [loading, setLoading] = useState(true);  
  
  const [error, setError] = useState(null);  
  
  const [search, setSearch] = useState("");  
  
  useEffect(() => {  
  
    const fetchData = async () => {  
  
      try {  
  
        const res = await fetch("https://jsonplaceholder.typicode.com/users");  
  
        if (!res.ok) throw new Error("Failed to fetch data");  
  
        const data = await res.json();  
  
        setStudents(data);  
  
      }  
  
      catch (err) {  
  
        setError(err.message);  
  
      }  
  
      finally {  
  
        setLoading(false);  
  
      }  
  
    }  
  
  }  
  
}
```



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```
    }  
  };  
  fetchData();  
  }, []);  
  const filteredStudents = students.filter((student) =>  
    student.name.toLowerCase().includes(search.toLowerCase())  
  );  
  return (  
    <div className="container">  
      <h1>Student Data Table</h1>  
      <input  
        type="text"  
        placeholder="Search by name..."  
        value={search}  
        onChange={(e) => setSearch(e.target.value)}  
      />  
      {loading && <div className="spinner"></div>}  
      {error && <p className="error">{error}</p>}  
  
      {!loading && !error && (  
        <StudentTable students={filteredStudents} />  
      )}  
    </div>  
  );  
}
```



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```
export default App;
```

- **StudentTable.jsx**

```
import React from "react";
```

```
function StudentTable({ students }) {
```

```
  return (
```

```
    <table>
```

```
      <thead>
```

```
        <tr>
```

```
          <th>Name</th>
```

```
          <th>Email</th>
```

```
          <th>Username</th>
```

```
          <th>City</th>
```

```
        </tr>
```

```
      </thead>
```

```
      <tbody>
```

```
        {students.map((student) => (
```

```
          <tr key={student.id}>
```

```
            <td>{student.name}</td>
```

```
            <td>{student.email}</td>
```

```
            <td>{student.username}</td>
```

```
            <td>{student.address.city}</td>
```

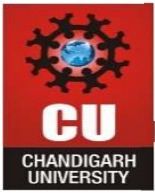
```
          </tr>
```

```
        )})
```

```
      </tbody>
```

```
    </table>
```

```
  );
```

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```
}
```

```
export default StudentTable;
```

4. Output:

Student Data Table			
Chel			
Name	Email	Username	City
Chelsey Dietrich	Lucio_Hettinger@annie.ca	Kamren	Roscoeview

5. Learning Outcome:

- **Understanding useEffect** for side-effects and API calls.
- **Ability to manage multiple states** (loading, error, data, search).
- **Skill in fetching and parsing JSON data** from APIs.
- **Confidence in error handling** for failed requests.
- **Capability to implement search filters** for dynamic data display.
- **Experience in building real-world React apps** with external data sources.